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17. Benacerraf's paper "What Numbers Could Not Be" ends with this paradoxical sounding passage: "[T]here are no such things as numbers; which is not to say that there are not at least two prime numbers between 15 and 20."[1]
What did the author mean?

Introduction

In his 1965 paper "What Numbers Could Not Be" Paul Benacerraf discusses a fundamental problem about the nature of numbers and the way we understand and refer to them in mathematical language. Mathematics is often seen as maximally precise and well-defined, yet questions about what numbers are reveal complexities at its very foundations. Benacerraf points out that set-theoretic models may capture the structure of arithmetic, but they fail to offer a satisfying account of what numbers actually are.. The argument culminates in a final sentence that appears to be paradoxical: "There are no such things as numbers; which is not to say that there are not at least two prime numbers between 15 and 20." This remark invites reflection on how arithmetic can be true and useful even if the number terms themselves do not correspond to particular objects.

To explain the meaning and philosophical significance of this line, I will proceed as follows: first, I will summarize each of the three sections of Benacerraf's paper, developing an understanding of how he builds his argument against identifying numbers with specific sets. Second, I will analyze the final sentence of the paper, showing how it uses rhetorical ambiguity to express Benacerraf's argument. Finally, I will explain how this conclusion anticipates and is foundational to the structuralist view in the philosophy of mathematics and explore its implications for how we should think about what number terms refer to and what kind of things, if any, numbers are.

[1]Benacerraf, Paul. "What Numbers Could Not Be." *The Philosophical Review*, vol. 74, no. 1, 1965, pp. 47–73. *JSTOR*, https://doi.org/10.2307/2183530.

Summary of Benacerraf

In Section I of "What Numbers Could Not Be" Benacerraf imagines a child named Ernie who was raised by parents who teach him about numbers through set theory rather than by the more traditional method of counting objects. Ernie learns that numbers are just certain sets, like the elements of an infinite set N structured by a recursive "less-than" relation R. For example, the number 1 is identified as the least element under R, and successors are defined in terms of R as well. As a result, Ernie does not need to separately learn the basic tenets of arithmetic. Rather, he can prove them from within the system he already understands.

Benacerraf describes how Ernie also learns to "count" in both intransitive and transitive senses: either by reciting numbers (intransitive), or by counting objects in a set by matching them one-to-one with the numbers starting from 1 (transitive). This lays the foundation for Ernie to understand cardinality; a set is said to have *k* elements if its members can be placed in one-to-one relation with the numbers from 1 up to k. For this notion to function within arithmetic, Benacerraf states that the ordering relation R must be recursive. That is, there must be a clear and finite procedure for determining which element comes next in the sequence. Without this, the number sequence could not be created and applied in practice. The section ultimately shows that the behavior of natural numbers can be understood entirely within carefully defined logical frameworks.

In Section II, Benacerraf introduces an important issue through comparing the method of Ernie to that of Johnny, another student taught arithmetic through set theory, however via a different framework. Although both frameworks involved recursive ordering, defined successor functions, and consistent accounts of cardinality, they assigned different sets to the same number terms. For example, Ernie defines 3 as {0,1,2}, while Johnny defines it as {{{0}}}. As a result, they disagree on seemingly factual statements, such as whether 3 is an element of 17.

This disagreement arises not from a difference in arithmetic knowledge but from a difference in representation. In Ernie's model, where each number is defined as the set of all

smaller numbers, 3 is literally one of the members of 17 since 17 is the set containing 0 through 16. In contrast, Johnny's model builds numbers through a series of successively nested sets, each containing the previous one. 3 is represented as a set with three layers of nesting and is entirely disjoint from 17. In Johnny's framework, 17 does not contain 3, or even any object that resembles 3 in Ernie's construction. The fact that two equally valid and internally consistent models lead to incompatible claims about identity demonstrates that number terms cannot precisely refer to any particular set. Their meaning must instead come from the structural role they play in their respective system. The identity of a number like 3, is thus not grounded in its set-theoretic composition, but in its position within a broader relational structure that governs how arithmetic functions and which statements are taken to be true.

This leads to a paradox: if both models meet all the requirements for being a valid theory of numbers, but assign different sets to define the same number, then numbers cannot *be* specific sets. After all, a number like 3 can't be two different things at once. Benacerraf argues that there is no non-arbitrary reason to prefer one identification over another, and no empirical or logical test that can decide which set the number 3 really *is*. Therefore, the practice of identifying numbers with particular sets is flawed. Benacerraf concludes that if number terms are tied to specific objects, the result is either inconsistencies between systems or an arbitrary choice of identification with one preferred over the other.

In Section III, Benacerraf proposes a way out of the dilemma by rejecting the assumption that numbers must be objects at all. He begins by critiquing Frege's idea that identity statements like "3 = [[[0]]]" must have a truth value. Benacerraf argues that this claim of identity, being between a number and a set, is meaningless unless the terms are part of the same conceptual framework with established rules for distinguishing one object from another. Since sets and numbers are individuated differently, treating 3 as identical to a specific set misunderstands the notion of identity. He then clarifies the distinction between modeling numbers and identifying them. The success of modeling arithmetic using recursive set

constructions does not prove that numbers *are* those sets. Rather, he argues that it only shows that certain structures can *replicate* number behavior.

Benacerraf concludes that numbers aren't objects with inherent identities but rather positions in a structure. Specifically, they are elements in any recursive progression. Arithmetic is the study of this structure, and not of particular objects. The individuality of numbers comes only from their relational positions, and not from intrinsic properties. Thus, the question "what is the number 3?" represents a misunderstanding of numbers. They are not objects, but rather positions in a system defined by recursive ordering.

The Final Line

Benacerraf ends this paper with the following line: "There are no such things as numbers; which is not to say that there are not at least two prime numbers between 15 and 20." This statement, while initially confusing, employs a rhetorical device called zeugma, using the term "numbers" according to two distinct senses within a single sentence. This rhetorical move illustrates Benacerraf's core thesis: we can reject numbers as being objects while still affirming the laws of arithmetic. The first clause of the sentence, "There are no such things as numbers," comes from the philosophical perspective expressed in the paper thus far: Benacerraf denies that numbers are objects with fixed identities, like sets or other abstract entities. Competing set-theory constructions like those of Ernie and Johnny each model arithmetic equally well, yet assign different sets to the same number. This suggests that numbers cannot be identified with any particular set. From this structuralist perspective, Benacerraf rejects the idea that the natural numbers form a unique set of objects, instead treating them as positions within any system that satisfies the appropriate recursive structure. Saying "2 exists" merely refers to the second position in such a structure, not to a metaphysically real entity.

The second clause of the sentence, "which is not to say that there are not at least two prime numbers between 15 and 20," shifts to the mathematical perspective, where arithmetic

continues to function even without an understanding of numbers as objects. Statements like that of the second clause remain true, regardless of the ontological status of numbers. This is because mathematical truths depend on the consistency and structure of arithmetic, not on whether numbers exist in a metaphysical sense.

By bridging these two uses of "number" with a rhetorical device, Benacerraf captures an important insight: we can preserve the practice and truth of mathematics without the assumption that numbers are objects. His conclusion reframes the question "What is the number 3?" as a category mistake, mistaking structural position for identity. Arithmetic, on this view, studies the relations between positions within recursive structures, not the nature of individual or freestanding entities.

Within Structuralism

Benacerraf's argument was key in the development of the philosophical theory of mathematical structuralism. His argument used the example of Ernie and Johnny to demonstrate that number terms cannot be understood as referring to particular objects, whether sets or any other entities, without introducing arbitrariness or contradiction. His account shifts the focus from what numbers are in and of themselves to how they are positioned within a relational structure. As Benacerraf puts it, numbers are not the kinds of things that "could have conquered Gaul"; they are not objects in any sense. This shift in focus became foundational for the development of structuralist accounts of mathematics.

Structuralism denies that there is a single, definitive natural number system. Instead, it holds that there are multiple equally valid systems that create the same structure. Correspondingly, statements about numbers are generalizations over any structure that satisfies the right formal conditions. To say that 2 exists is simply to refer to "the thing in the second place" in some recursively ordered progression. According to this view, arithmetic studies the structure itself, how elements relate through recursion, succession, and order, but not the intrinsic nature of any one such possible system. Benacerraf's paper played a central role in

advancing this shift. His identification problem showed that multiple, equally valid set-theoretic models can capture the structure of the natural numbers, thus undermining the idea that numbers are any particular things, and reinforcing the structuralist claim that it is the pattern, not the substance, that matters.

Conclusion

Benacerraf's paper "What Numbers Could Not Be" challenges the assumption that numbers must be understood as objects in order for arithmetic to be true and meaningful. By showing that multiple, contradictory set-theoretic constructions can just as well model the structure of the natural numbers, he raises a serious problem for any view that seeks to identify numbers with particular entities. His resolution, which lays the groundwork for the structuralist approach, reframes numbers as positions in a system rather than objects with intrinsic identity. The use of zeugma in the paper's final sentence captures this distinction between an ontological and structural understanding of numbers: while there may be no such things as numbers as concrete or identifiable objects, arithmetic statements involving numbers can still be true when understood structurally. Benacerraf's argument thus shifts attention from the metaphysical status of numbers to the system of rules and relations that give arithmetic its structure, offering a framework in which mathematical practice can proceed without assuming that numbers are objects.